# CS 486/686
# Pretraining Language Models From First Principles

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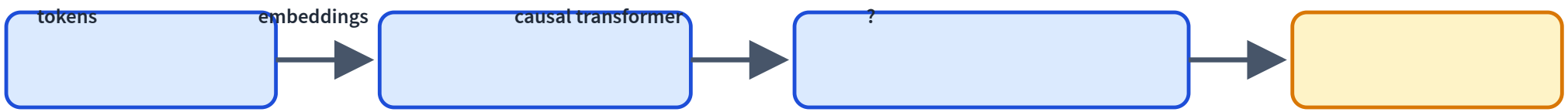
Lecture 20

How a transformer learns to predict text

# Learning goals

- Turn **text** into tokens and embeddings.
- Explain **next-token prediction** and **cross-entropy**.
- Explain **pretraining** as self-supervised learning at scale.
- Distinguish **training** from **decoding** / **inference**.

# From architecture to objective




Architecture alone does nothing. We need a training signal.

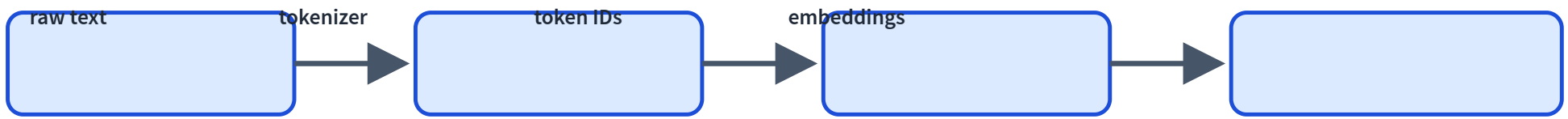
# The language modeling task

The robot picked up the cup because it was ___

| **empty** | **blue** | **quantum** |
| --- | --- | --- |
| likely | possible | unlikely |

Language modeling means predicting what comes next.

# Text is not directly a tensor




Models see token IDs and vectors, not words as humans see them.

# Why tokenization is hard

`cat` `cats` `unbelievable` `CS486` `spaces?`

**word vocab**

brittle for rare words

**character vocab**

very long sequences

# Subwords are the compromise

Common pieces become reusable tokens.

`unbelievable`

**un** **believe** **able**

Exact splits depend on the tokenizer, but the idea is the same: balance vocabulary size and sequence length.

# Tokenizer playground LIVE

See how real text splits into subword tokens and IDs.

A tokenizer splits text into **subword** pieces, each with an integer id. The dot (·) marks a leading space.

"unbelievable" | "The robot picked up the cup." | "CS486 tokenization isn't trivial" | "transformers"

unbelievable

tokenize

| un | bel | iev | able |
| --- | --- | --- | --- |
| 403 | 6667 | 11203 | 540 |

12 characters

4 tokens

Common words stay whole; rare words split into reusable pieces (un·bel·iev·able).

# Token IDs become embeddings

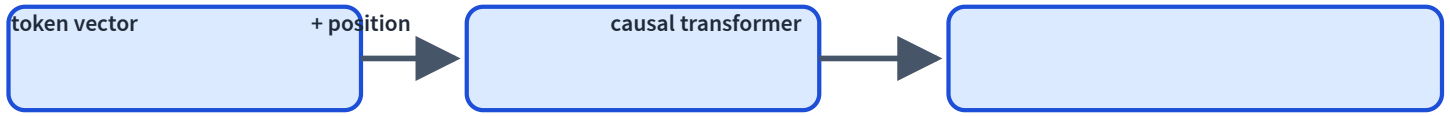
| token | id | embedding row |
|---|---|---|
| The | 791 | [0.12, -0.40, ...] |
| robot | 21933 | [0.51, 0.08, ...] |
| cup | 7721 | [-0.20, 0.33, ...] |

This is the embedding table from L18.

Pretraining updates these rows.

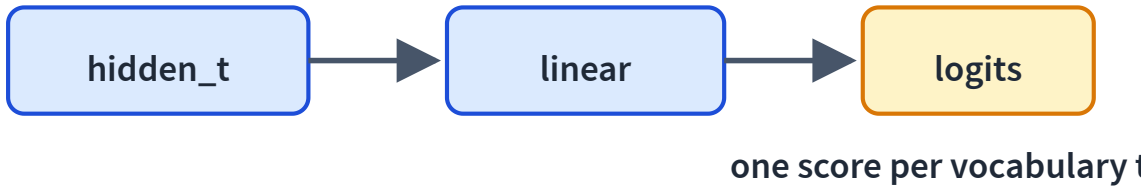
# Add position, then transformer blocks

$$\text{input}_t = \text{token\_embedding}_t + \text{position}_t$$




Each position can only use earlier context because of the causal mask.

# The output head predicts a token




At every position, the model outputs a vector of scores.

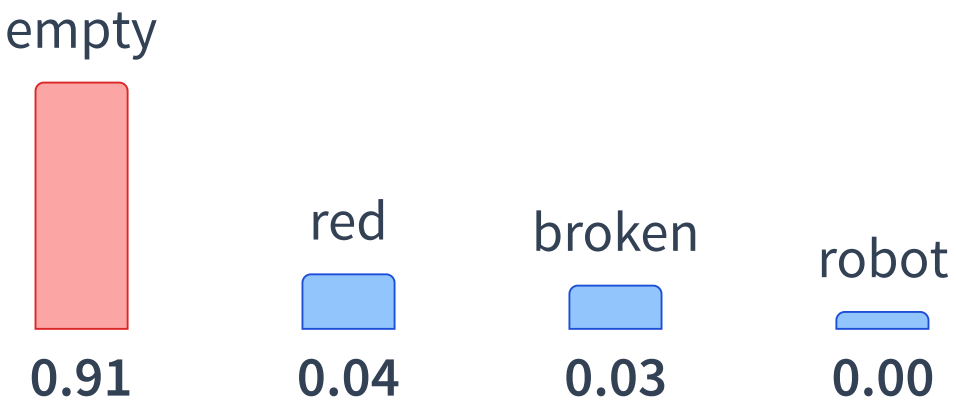
If the vocabulary has 150k tokens, there are 150k logits.

# Logits are scores, not probabilities

| candidate next token | logit |
|---|---|
| **empty** | **8.2** |
| red | 3.1 |
| broken | 2.7 |
| robot | -1.4 |

Bigger logit = more preferred, but not yet a probability.

# Softmax turns logits into probabilities



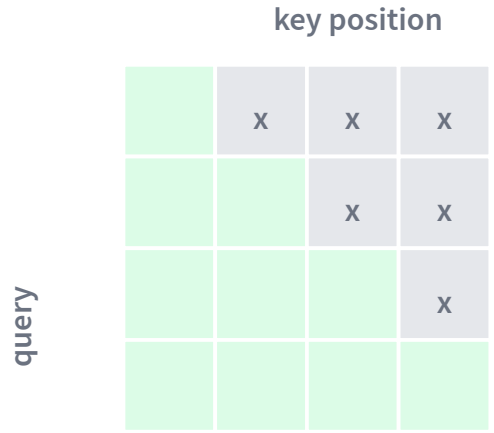

Now we have a next-token distribution.

# One sequence gives many examples

Raw text supervises itself by shifting one token right.

| input context | target |
|---|---|
| `The` | **robot** |
| `The robot` | **picked** |
| `The robot picked` | **up** |
| `The robot picked up the` | **cup** |

# Causal masking makes training legal




Position $t$ predicts token $t + 1$.

It may only see tokens $\leq t$.

No future peeking, even during training.

# Autoregressive factorization

A text probability is built one next-token prediction at a time:

$$P(x_1, \ldots, x_T) = \prod_{t=1}^{T} P(x_t \mid x_{<t})$$

Same idea as generation: left to right, token by token.

## Cross-entropy for the next token

If the true next token is **empty**:

$$\text{loss} = -\log P(\text{empty} \mid \text{context})$$

This is multiclass classification over the vocabulary.

# The loss, token by token LIVE MODEL

Each bar is $-\log p(\text{actual next token})$ — one sentence, many signals.

The robot picked up the cup becau

score my own text (real model)

·robot 15.3
·picked 8.2
·up 0.4
·the 1.5
·cup 7.9
·because 6.8
·it 1.1
·was 1.2
·empty 6.1

average loss = 5.37 nats/token
low = predictable (“up” after “picked”); high = surprising

# The pretraining loop

```
ids = tokenizer(text)
x = ids[:-1]
y = ids[1:]

logits = model(x)                # [T, vocab_size]
loss = cross_entropy(logits, y)
loss.backward()
optimizer.step()
```

This is L17's training loop, scaled to billions/trillions of tokens.

# What gets updated?

**embeddings**

token vectors

**attention**

retrieval weights

**feed-forward**

per-token compute

**output head**

logits

Pretraining changes all model parameters.

# Why this is self-supervised

No human labels are required.

Raw text gives both inputs and targets by shifting tokens.

Scale becomes possible because labels are essentially free.

But data quality still matters.

# What does pretraining teach?

**patterns**

grammar, style, syntax

**associations**

facts seen in data

**skills**

traces from
text and code

It learns statistical prediction, not guaranteed truth.

# Pretraining at scale


Modern LMs improve when these are scaled carefully together.

# Training is not inference

## Training

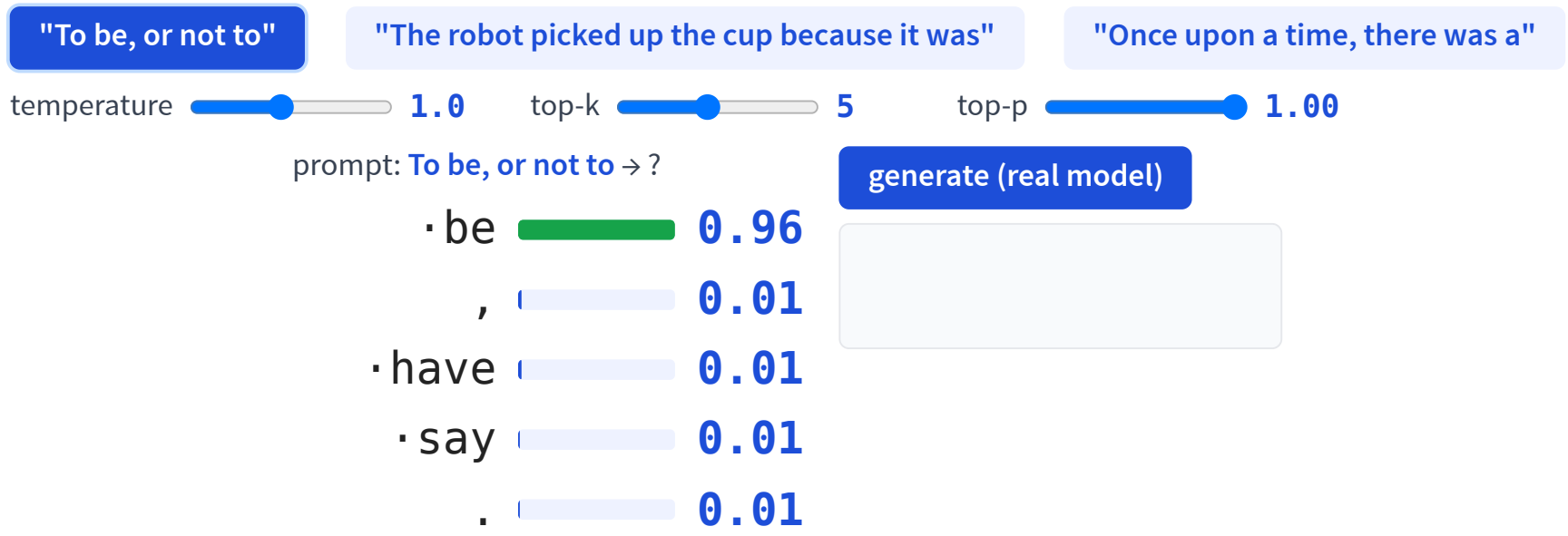
many tokens in parallel
targets are known
weights update

## Inference

one new token at a time
no target label
weights fixed

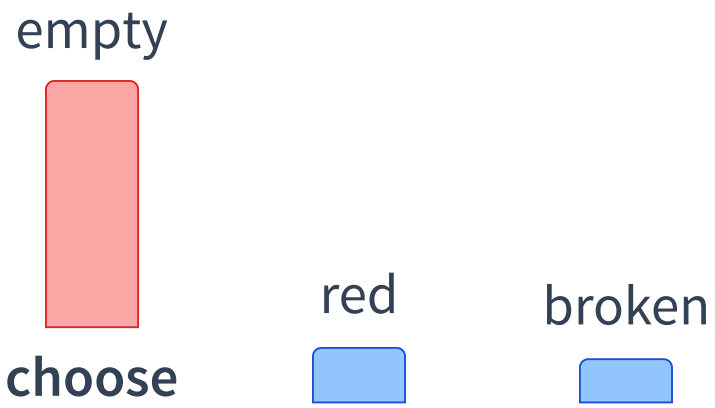
# Decoding: from distribution to text LIVE MODEL

A real next-token distribution; the knobs reshape it, then generate.



# Greedy decoding

Pick the highest-probability token each step.




Simple and stable, but often dull or brittle.

# Sampling and temperature

Temperature reshapes the distribution:

$$\mathrm{softmax}(\mathrm{logits}/T)$$

**low $T$**

sharper, safer, repetitive

**high $T$**

flatter, diverse, risky

# Temperature, hands-on LIVE

Low temperature sharpens onto one token; high temperature flattens toward uniform.

temperature 1.0

dog 1.0 cat 3.0 bird 1.5 fish 0.5


Softmax turns scores into probabilities. Low temperature sharpens; high temperature flattens (the decoding knob in L20/L21).

# Restrict randomness to plausible tokens

**top-k**

sample from the $k$ most likely tokens

**top-p**

sample from the smallest set with total probability $\geq p$

Decoding choices shape output even when weights are fixed.

# What pretraining does not solve

### truth

fluent text
can be wrong

### instructions

base models
complete text

### private data

not known
unless supplied

Later: prompting, fine-tuning, RAG, and tools.

# Next: open the hood

Now we know:

- the architecture: causal transformer
- the objective: next-token prediction
- the inference step: sample the next token

L21: Dissecting a Small Language Model.

# Recap

✓ Text becomes **tokens, IDs, embeddings**.

✓ A causal transformer predicts **next-token logits**.

✓ **Softmax + cross-entropy** train the model.

✓ Pretraining is **self-supervised** at scale.

✓ Decoding turns probabilities into generated text.